Aprendizagem Aplicada à Segurança

Aprendizagem Aplicada à Segurança

Mário Antunes

September 18, 2026

Universidade de Aveiro

Professor

- **Name:** Mário Antunes
- **E-Mail:** mario.antunes@ua.pt
- **Office:** 19.2.15 (IT1)

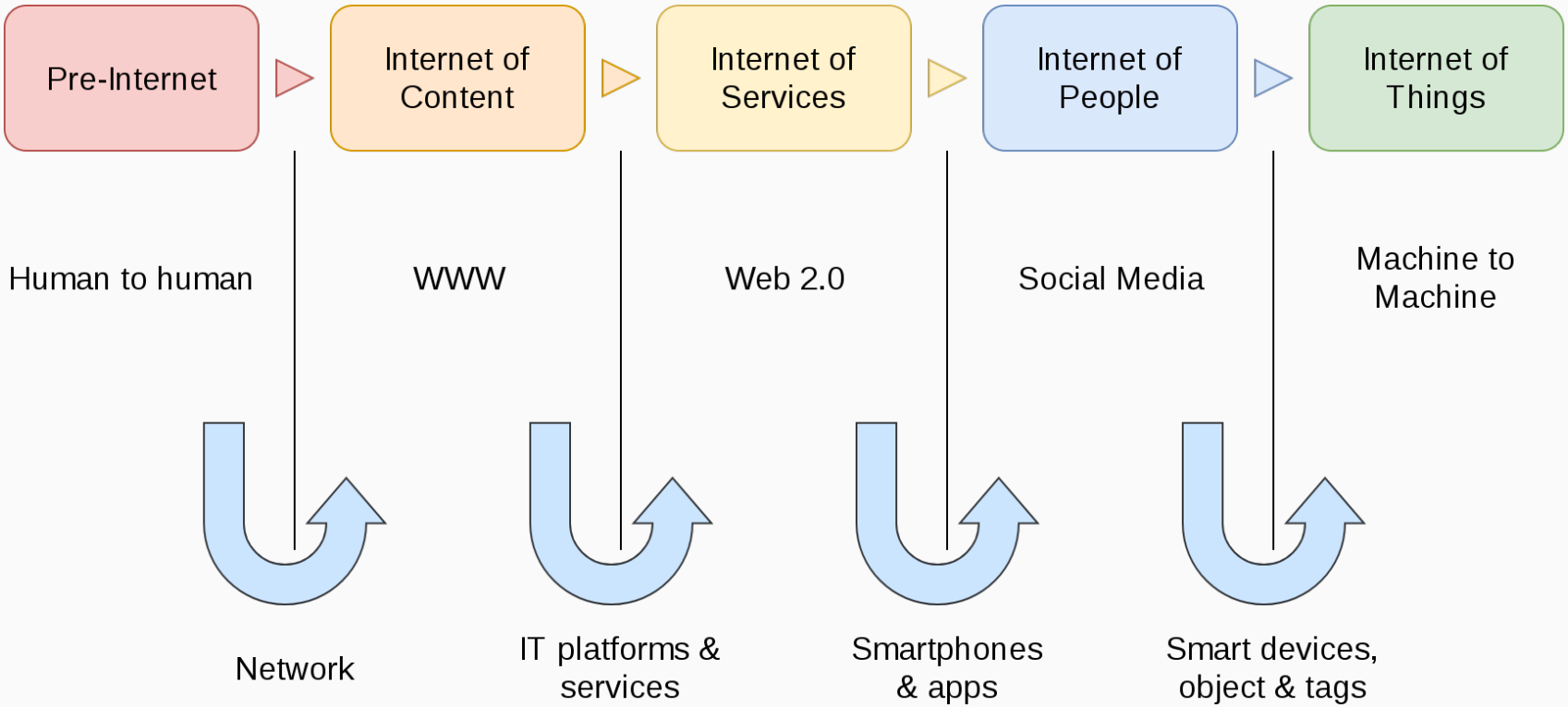


Class Introduction i

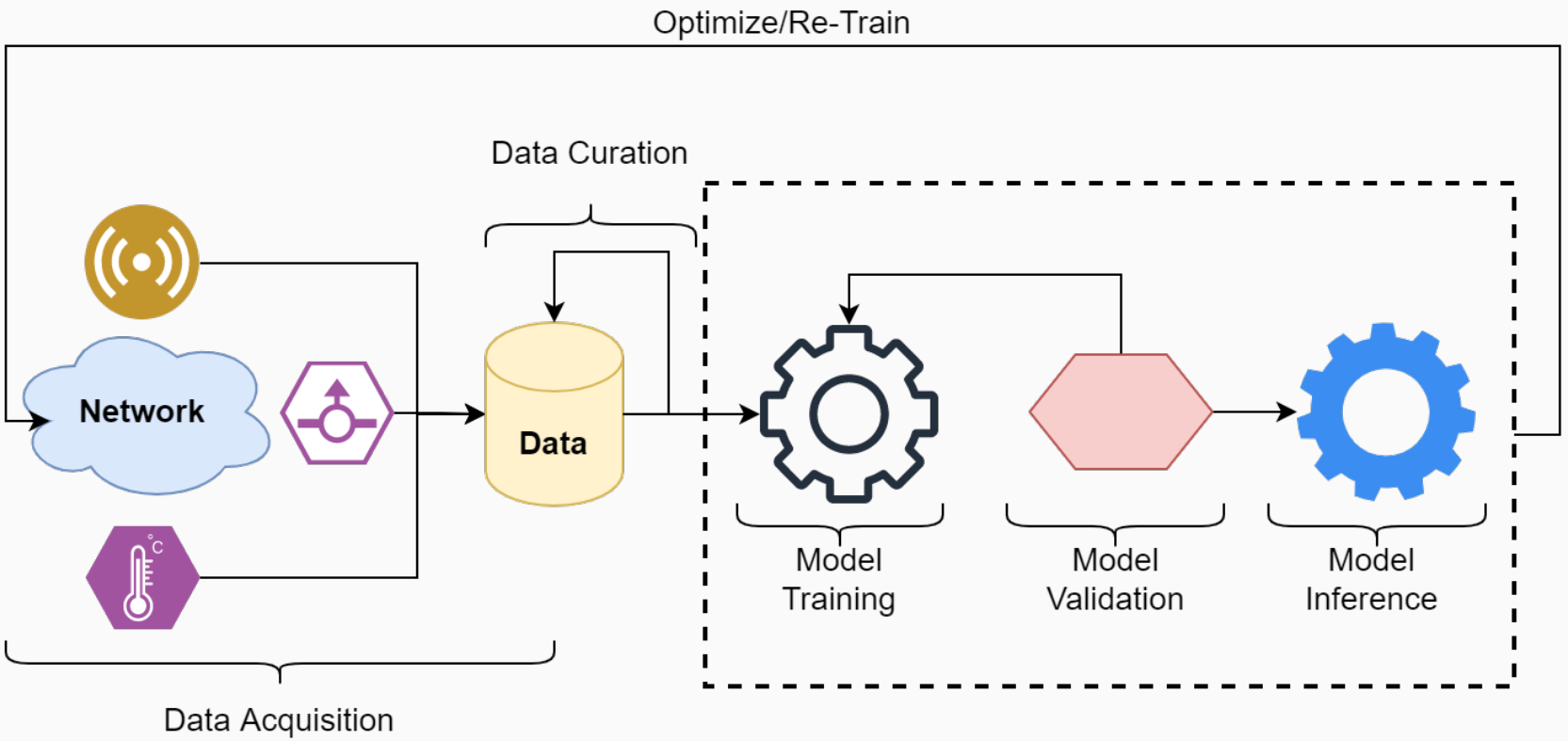
- Given the evolution of the threats
- And the complexity of the systems
- AI/ML are gaining traction as a usefull tool



Class Introduction ii



Class Introduction iii



Evaluation

- **Discrete (recommended):** 50% theory + 50% practice
  - Theoretical Test 1 (25%): Class 08, 06-11-2026 (Modules 1 and 2: spam and anomaly detection)
  - Theoretical Test 2 (25%): Class 14, 18-12-2026 (Modules 3 and 4)
  - Practical project (50%): released Class 08 (06-11-2026), due Class 14 (18-12-2026)
- **Final:** 50% final exam (*Época Normal*) + 50% project

Class Schedule

Date	Class	Topic
18-09-2026	1	Presentation, AI/ML for security
25-09-2026	2	Spam detection I
02-10-2026	3	Spam detection II
09-10-2026	4	Spam detection III
16-10-2026	5	Anomaly detection I
23-10-2026	6	Anomaly detection II
30-10-2026	7	Anomaly detection III
06-11-2026	8	Test 1 + project kickoff
13-11-2026	9	Malware analysis I
20-11-2026	10	Malware analysis II
27-11-2026	11	Network intrusion detection
04-12-2026	12	Generative AI and LLMs in security
11-12-2026	13	Project clinic
18-12-2026	14	Test 2 + project presentations

Environment

























Bibliography i

All of the books are available here:
https://learning.oreilly.com/

Chio, Clarence, and David Freeman. 2018. *Machine Learning and Security*. O'Reilly.

Halder, Soma, and Sinan Ozdemir. 2018. *Hands-on Machine Learning for Cybersecurity: Safeguard Your System by Making Your Machines Intelligent Using the Python Ecosystem*. Packt Publishing Ltd.

Bibliography ii

Mueller, John Paul, and Rod Stephens. 2019. *Machine Learning Security Principles*. Packt Publishing Ltd.

Parisi, Alessandro. 2019. *Hands-on Artificial Intelligence for Cybersecurity: Implement Smart AI Systems for Preventing Cyber Attacks and Detecting Threats and Network Anomalies*. Packt Publishing Ltd.

Tsukerman, Emmanuel. 2019. *Machine Learning for Cybersecurity Cookbook*. Packt Publishing Ltd.